COMP108
Data Structures and Algorithms

Divide-and-Conquer Algorithms (Part II Merge Sort Algorithm)

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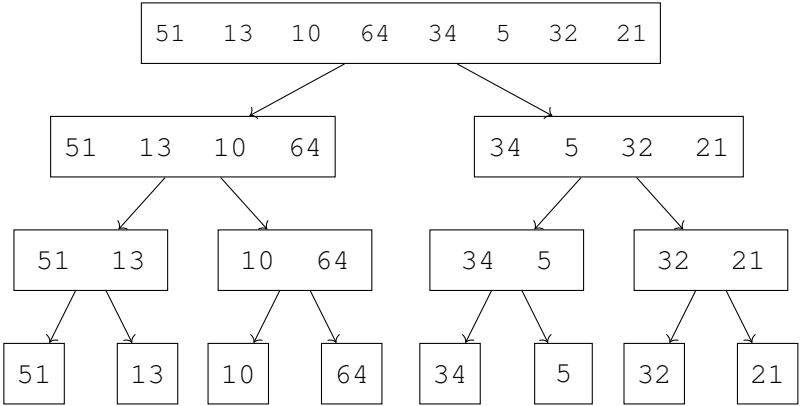
Merge Sort

- using divide and conquer technique
- divide the sequence of n numbers into two halves
- recursively sort the two halves
- merge the two sorted halves into a single sorted sequence
  - It is easier to merge two sequences that are already sorted, compared to two non-sorted sequences.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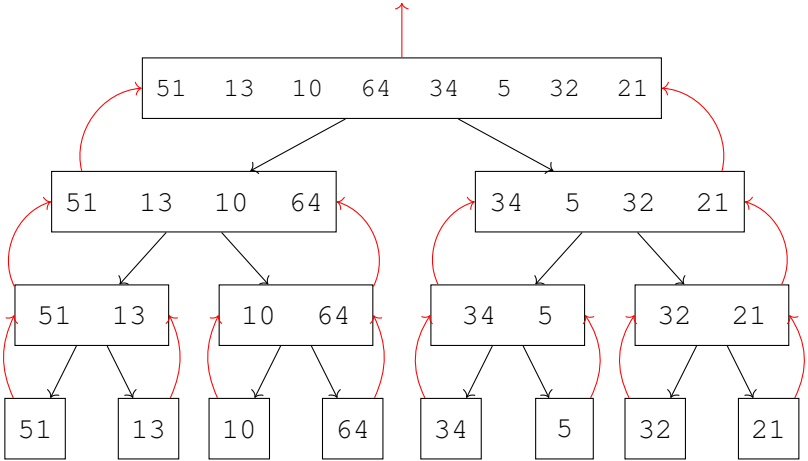
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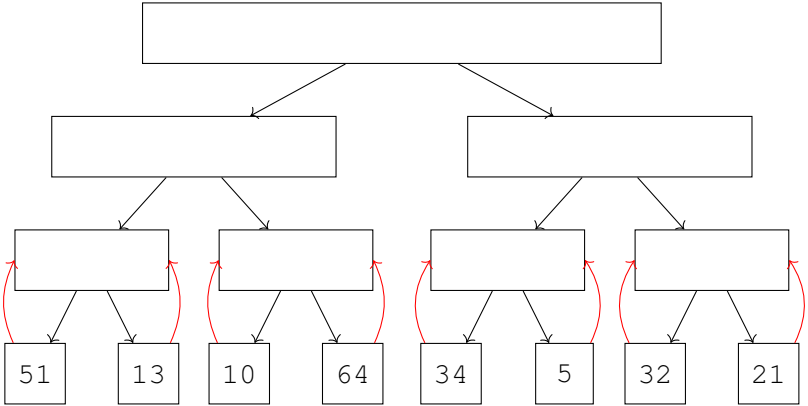
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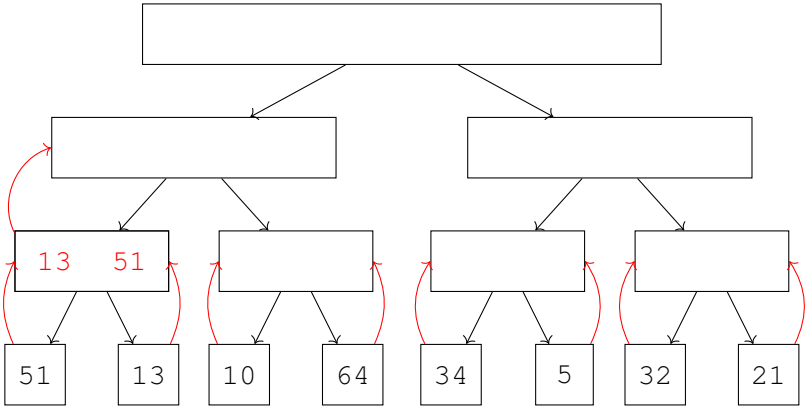
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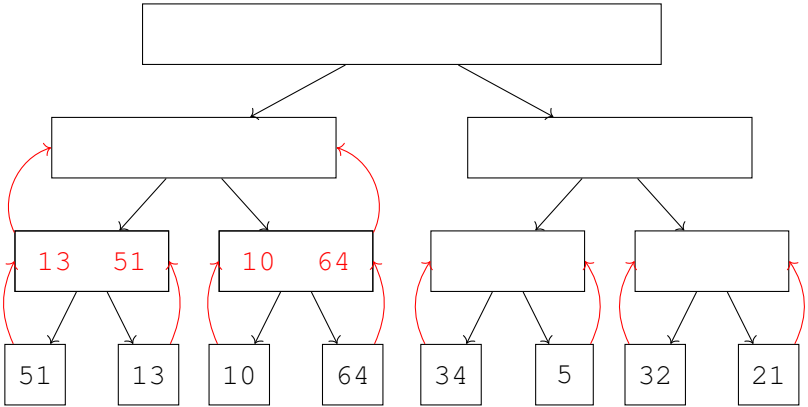
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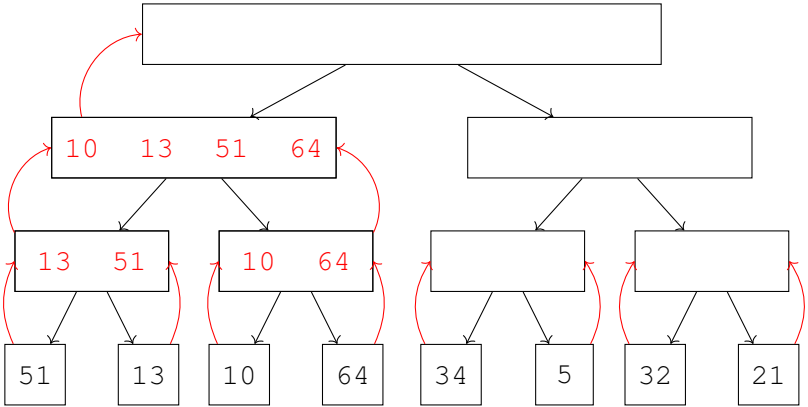
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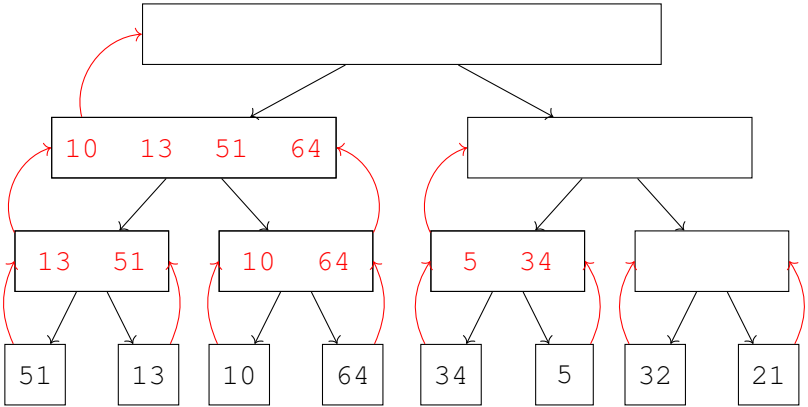
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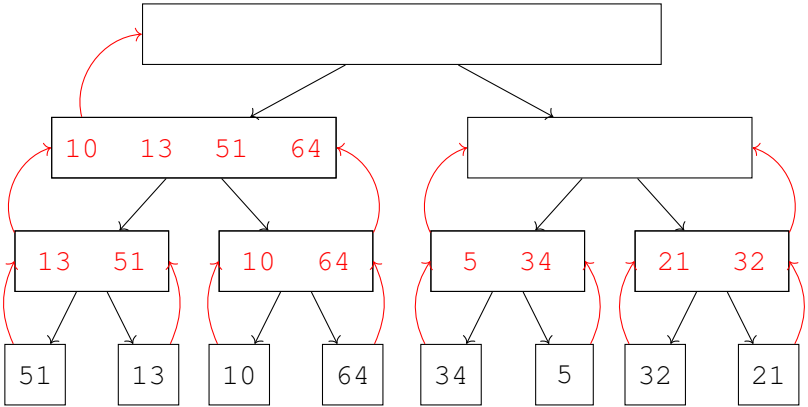
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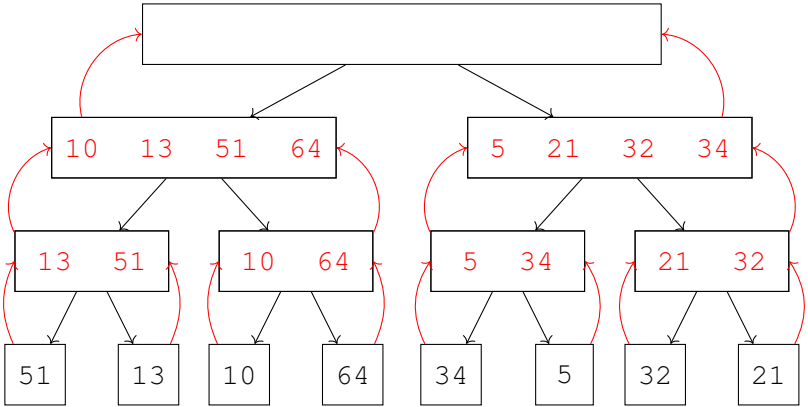
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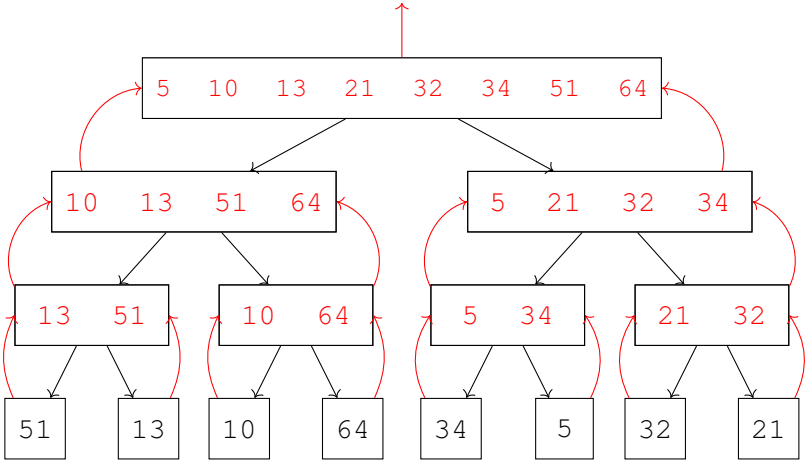
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Merge Sort Algorithm SampleMergeSort.java

For simplicity, assume n is a power of 2

Algorithm MergeSort($A[1..n]$)

if $n > 1$ then

begin

end

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if $n > 1$ then

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copy $A[1..\frac{n}{2}]$ to $B[1..\frac{n}{2}]$

copy $A[(\frac{n}{2}+1)..n]$ to $C[1..\frac{n}{2}]$

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MergSort(C)

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Merge(B, C, A)

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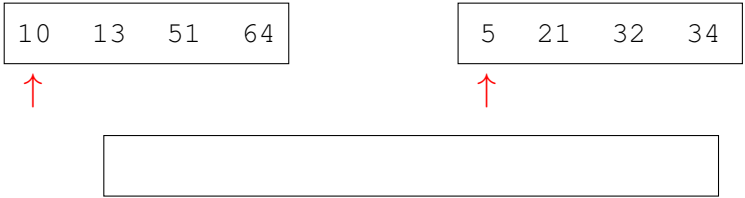
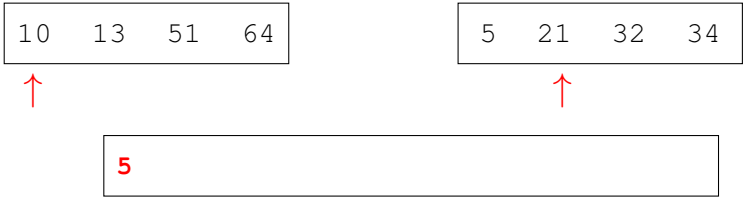
How to merge?



To merge two sorted sequences, we keep two **pointers**, one to each sequence

Compare the two numbers pointed, copy the **smaller** one to the result and **advance** the corresponding pointer



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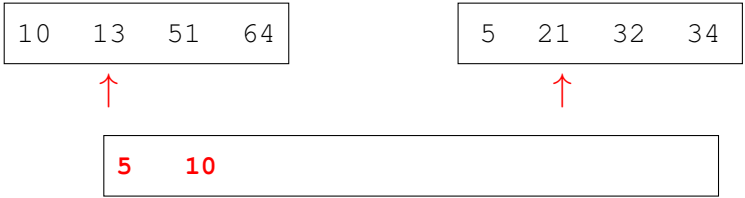
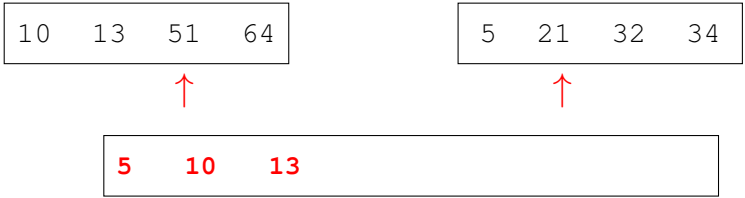


Repeat the process · · ·



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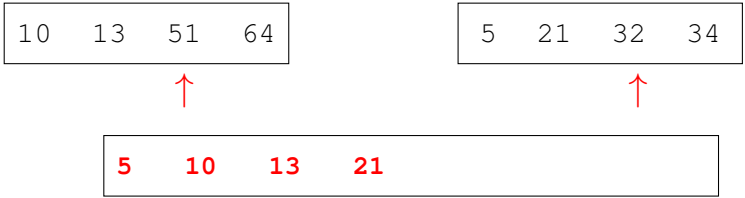


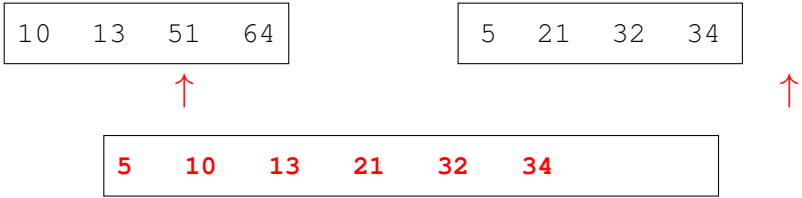
Repeat the process · · ·

10	13	51	64

5	21	32	34

5	**10**	**13**	**21**	**32**

Repeat the process · · ·



When we reach the **end** of one sequence, simply copy the **remaining** numbers in the other sequence to the result

10	13	51	64

5	21	32	34

5	**10**	**13**	**21**	**32**	**34**	**51**	**64**

Then we obtain the final sorted sequence

Merge Algorithm

Algorithm Merge($B[1..p]$, $C[1..q]$, $A[1..(p+q)]$)

$i \leftarrow 1, \quad j \leftarrow 1, \quad k \leftarrow 1$

Merge Algorithm

Algorithm Merge($B[1..p]$, $C[1..q]$, $A[1..(p+q)]$)

$i \leftarrow 1, \quad j \leftarrow 1, \quad k \leftarrow 1$

while $i \leq p$ AND $j \leq q$ do

begin

$k \leftarrow k + 1$

end

Merge Algorithm

Algorithm Merge($B[1..p]$, $C[1..q]$, $A[1..(p+q)]$)

$i \leftarrow 1, \quad j \leftarrow 1, \quad k \leftarrow 1$

while $i \leq p$ AND $j \leq q$ do

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if $B[i] \leq C[j]$ then

else

$k \leftarrow k + 1$

end

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$i \leftarrow 1, \quad j \leftarrow 1, \quad k \leftarrow 1$

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if $B[i] \leq C[j]$ then

$\boldsymbol{A[k] \leftarrow B[i], \quad i \leftarrow i+1}$

else

$k \leftarrow k+1$

end

Merge Algorithm

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Algorithm Merge(B[1..p], C[1..q], A[1..(p + q)])
    i ← 1,   j ← 1,   k ← 1
    while i ≤ p AND j ≤ q do
    begin
        if B[i] ≤ C[j] then
            A[k] ← B[i],   i ← i + 1
        else
            A[k] ← C[j],   j ← j + 1
        k ← k + 1
    end
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else

$\mathbf{A[k] \leftarrow C[j], \quad j \leftarrow j+1}$

$k \leftarrow k+1$

end

if $i == p+1$ then

else

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```
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    i ← 1,   j ← 1,   k ← 1
    while i ≤ p AND j ≤ q do
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        else
            A[k] ← C[j],   j ← j + 1
        k ← k + 1
    end
    if i == p + 1 then
        copy C[j..q] to A[k..(p + q)]
    else
```

Merge Algorithm

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    i ← 1,   j ← 1,   k ← 1
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    end
    if i == p + 1 then
        copy C[j..q] to A[k..(p + q)]
    else
        copy B[i..p] to A[k..(p + q)]
```

Time complexity?

Merge Algorithm

p=4

B: | 10 | 13 | 51 | 64 |

q=4

C: | 5 | 21 | 32 | 34 |

	i	j	k	$A[\,]$
before loop	1	1	1	empty

Merge Algorithm

p=4

B: | 10 | 13 | 51 | 64 |

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C: | 5 | 21 | 32 | 34 |

	i	j	k	$A[\,]$
before loop	1	1	1	empty
end of 1st iteration	1	2	2	5

Merge Algorithm

p=4

B: | 10 | 13 | 51 | 64 |

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	i	j	k	$A[\,]$
before loop	1	1	1	empty
end of 1st iteration	1	2	2	5
end of 2nd iteration	2	2	3	5 10

Merge Algorithm

p=4

B: | 10 | 13 | 51 | 64 |

q=4

C: | 5 | 21 | 32 | 34 |

	i	j	k	$A[\,]$
before loop	1	1	1	empty
end of 1st iteration	1	2	2	5
end of 2nd iteration	2	2	3	5 10
end of 3rd	3	2	4	5 10 13

Merge Algorithm

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B: | 10 | 13 | 51 | 64 |

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	i	j	k	$A[\,]$
before loop	1	1	1	empty
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end of 4th	3	3	5	5 10 13 21

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end of 4th	3	3	5	5 10 13 21
end of 5th	3	4	6	5 10 13 21 32

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end of 4th	3	3	5	5 10 13 21
end of 5th	3	4	6	5 10 13 21 32
end of 6th	3	5	7	5 10 13 21 32 34

Merge Algorithm

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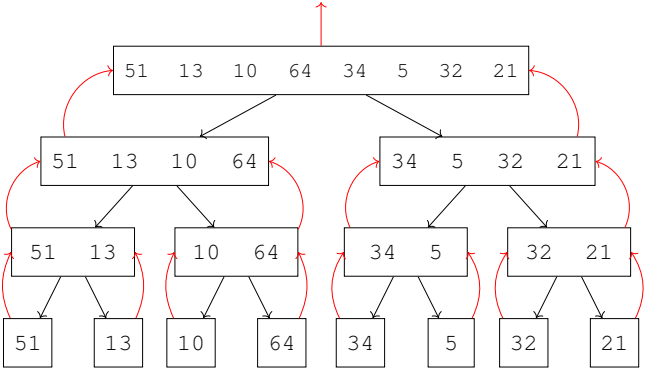
C: | 5 | 21 | 32 | 34 |

	i	j	k	$A[\,]$
before loop	1	1	1	empty
end of 1st iteration	1	2	2	5
end of 2nd iteration	2	2	3	5 10
end of 3rd	3	2	4	5 10 13
end of 4th	3	3	5	5 10 13 21
end of 5th	3	4	6	5 10 13 21 32
end of 6th	3	5	7	5 10 13 21 32 34
after if-else				5 10 13 21 32 34 51 64

Time complexity analysis



- Each node takes $O(r)$ time when there are r integers.
- Each level takes $O(n)$ time because total number of integers is n.
- There are $O(\log n)$ levels.
- Overall: $\boldsymbol{O(n \log n)}$ time.

Summary

Summary: Merge Sort Algorithm

Next: Fibonacci Numbers

For note taking